VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI – 590 018,

KARNATAKA

Shri Bhagwan Mahaveer Jain Educational & Cultural Trust ®

JAIN COLLEGE OF ENGINEERING, BELAGAVI

Estd. 2010

DEPARTMENT OF ELECTRONICS AND COMMUNICATIONENGINEERING

FOURTH YEAR

(2025 – 2026)

PROJECT REPORT

On

"AI powered smart multipurpose agribot"

PROJECT GUIDE

Mr.Viraj.J

Professor, Dept. of E&CE

JCE,Belagavi

PROJECT MEMBER

Mr. Abhishek.D	2JI22EC002
Mr. Amit .K	2JI22EC013
Mrs. Kavya.K	2JI22EC053
Mrs. Manisha.A	2JI22EC062

Shri Bhagwan Mahaveer Jain Educational & Cultural Trust ®

JAIN COLLEGE OF ENGINEERING, BELAGAVI

T.S.NAGAR, MACHHE – 590014, KARNATAKA

Estd. 2010

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the **Project Work** entitled **"AI powered smart multipurpose agribot"** carried out by **Mr.Abhishek.D (2JI22EC002),Mr.Amit.K(2JI22EC013),Mrs.Manisha(2JI22EC062),Mrs.Kavya.K(2JI22EC053)** are bonafide students of **Department of Electronics and Communication Engineering, Jain College of Engineering, Belagavi,** in partial fulfilment for the award of **Bachelor of Engineering** of the **Visvesvaraya Technological University, Belagavi** during the academic year **2025-2026**. It is certified that all corrections/suggestions indicated for project assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

Prof. Viraj.J	**Prof. V R Bagali**	**Dr. J. Shivakumar**
Project Guide	HOD, Dept. of E&CE	Principal & Director
Dept. of E&CE, JCEBGM	JCE, Belagavi	JCE, Belagavi

Name of the Evaluators Signature with date

1. ____________________________ ____________________

2. ____________________________ ____________________

DECLARATION

We, **Mr. Abhishek D (2JI22EC002), Mr. Amit K (2JI22EC013), Mrs. Manisha (2JI22EC062), and Mrs. Kavya K (2JI22EC053)**, students of **7th Semester, B.E. Electronics & Communication Engineering**, **Jain College of Engineering, Belagavi**, hereby declare that the dissertation entitled **"Ai powered smart multipurpose agribot"** has been carried out as a batch project and is submitted in partial fulfilment of the requirements of the degree of **Bachelor of Engineering in Electronics & Communication Engineering** under **Visvesvaraya Technological University (VTU), Belagavi**, during the academic year **2025–2026**.

Name	USN	Signature
Mr. Abhishek.D	2JI22EC002	______________________
Mr. Amit.K	2JI22EC013	______________________
Mrs. Kavya.K	2JI22EC053	______________________
Mrs. Manisha.A	2JI22EC062	______________________

Place : Belagavi

Date :

Vision of Institute

"To be a university as a resource of solution to diverse challenges of society by nurturing innovation, research & entrepreneurship through value based education."

Mission of Institute

- To provide work culture that facilitates effective teaching-learning process and lifelong learning skills.
- To promote innovation, collaboration and leadership through best practices.
- To foster industry-institute interaction resulting in entrepreneurship skills and employment opportunities.

Vision of Department

"To achieve excellence in education and research for developing globally competent, ethically sound Electronics & Communication Engineers".

Mission of Department

1. To provide conducive environment through structured student centric, teaching learning process.
2. To nurture needs of society by infusing scientific temper in students and to grow as a centre of excellence with efficient industry-institute interaction.
3. To inculcate self-learning skills, entrepreneurial ability, and professional ethics.

Program Educational Objectives (PEO's)

1. Graduates will be able to contemplate real-time social problems and deliver efficient solutions.
2. Graduates will be able to lead and succeed in professional careers.
3. Graduates will contribute through research and entrepreneurship.

Program Specific Outcomes (PSO's)

Graduates in the UG program in Electronics and communication engineering will be able to

1. Design, verify and develop analog and digital systems by using state of art technology to contribute to the societal needs.
2. Apply knowledge in various domain of IoT, real time systems, communication systems, VLSI and embedded systems, image and signal processing using hardware and software tools.

PROGRAM OUTCOME'S (PO'S)

Engineering Graduates will be able to:

1. **Engineering knowledge**: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis**: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions**: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems**: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage**: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society**: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability**: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics**: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication**: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance**: Demonstrate knowledge and understanding of the **engineering and management principles and apply these to one's own work, as a member and** leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning**: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

CO-PO/PSO Mapping:

L1: Remembering L2: Understanding L3: Applying L4: Analyzing L5: Evaluating L6: Creating

Course Outcomes	Description	Bloom's Cognitive level
22BEC786.01	**Understand the basic concepts and broad principles of industrial projects.**	**L3**
22BEC786.02	**Apply the theoretical concepts to solve industrial problems with team work and multidisciplinary approach.**	**L3**
22BEC786.03	**Demonstrate professionalism with ethics. Present effective communication skills related to engineering issues to broader societal context.**	**L3**
22BEC786.04	**Get capable of self-education and clearly understand the values of achieving perfection in project implementation and completion.**	**L3**
22BEC786.05	**Understand concepts of projects and production management.**	**L3**

Strength of CO Mapping to PO/PSOs with Justification:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	1		1	1							3
CO2	3	3	3	1					3				3	3
CO3								2		3				
CO4		3	3		2		2	1	3	2	3	2	2	2
CO5									3	3	2	2	3	
Avg	3	3	3	1		1	1	2	3	3	2	2	3	3

CO-PO-PSO	Justification
CO1 → PO1 (3) CO1 → PO2 (3) CO1 → PO3 (3) CO1 → PO4 (1) CO1 →PO6(1) CO1 →PO7(1) CO1 → PSO3(2)	• Strongly mapped as students apply the electronics and communication engineering skills to understand the principles of industrial projects. • Strongly mapped as students identify, formulate, review research literature and analyse complex engineering problems. • Strongly mapped as students design solutions for complex engineering problems with appropriate consideration for the public health and safety. • Slightly mapped as students can use research-based knowledge and methods including design of experiments, analysis and interpretation to provide valid conclusions. • Slightly mapped as students can able to understand the impact of professional engineering project solutions in societal and environmental contexts. • Slightly mapped as students can be able to apply reasoning of contextual project knowledge to assess societal, health and safety issues. • Strongly mapped as apply knowledge in various domain of IoT, real time systems, communication systems, VLSI and embedded systems, image and signal processing using hardware and software tools.
CO2→ PO1 (3) CO2→ PO2 (3) CO2→ PO3 (3) CO2→ PO4 (1) CO2→ PO9 (3) CO2→ PSO1(3) CO2→ PSO2(3)	• Strongly mapped as students apply the electronics and communication engineering skills to understand the principles of industrial projects. • Strongly mapped as students identify, formulate, review research literature and analyse complex engineering problems. • Strongly mapped as students design solutions for complex engineering problems with appropriate consideration for the public health and safety. • Slightly mapped as students can use research-based knowledge and methods including design of experiments, analysis and interpretation to provide valid conclusions. • Strongly mapped as students can effectively act as an individual, and as a member or leader and work in a team. • Strongly mapped as students can design, verify and develop analog and digital systems by using state of art technology to contribute to the societal needs and solve industrial problem. • Strongly mapped as students apply knowledge in various domain of IoT, real time systems, communication systems, VLSI and embedded systems, image and signal processing using hardware and software tools and solve industrial problem.

CO3→ PO8 (2) CO3→ PO10 (3)	• Moderately mapped as students can work ethically and professionally in the industry. • Strongly mapped as Students can comprehend and write effective reports and make documentation with effective presentation.
CO4→ PO2 (3) CO4→ PO3 (3) CO4→ PO5 (2) CO4→ PO12 (2) CO4→ PSO1(2)	• Strongly mapped as Students can effectively act as an individual, and as a member or leader and work in a team. • Strongly mapped as Students can comprehend and write effective reports and make documentation with effective presentation. • Moderately mapped as students can learn to management and financial skills required for the execution of project. • Moderately mapped as students can be engaged in lifelong learning in the broadest context of technological change through project implementation and completion. • Moderately mapped as students can design, verify and develop analog and digital systems by using state of art technology to contribute to the societal needs and solve industrial problem.
CO5→ PO9 (3) CO5→ PO10 (3) CO5→ PO11 (2) CO5→ PO12 (2) CO5→ PSO1(3)	• Strongly mapped as Students can effectively act as an individual, and as a member or leader and work in a team. • Strongly mapped as Students can comprehend and write effective reports and make documentation with effective presentation. • Moderately mapped as students can learn to management and financial skills required for the execution of project. • Moderately mapped as students can be engaged in life long learning in the broadest context of technological change through project implementation and completion. • Strongly mapped as students can design, verify and develop analog and digital systems by using state of art technology to contribute to the societal needs and solve industrial problem. •

ACKNOWLEDGEMENTS

Although a single sentence hardly suffices, we would like to thank almighty God for blessing us with his grace and taking our endeavour to a successful culmination.

We express our gratitude to our guide **Guide Name, Professor, Dept. of E&CE, JCE, Belagavi,** for his valuable guidance and continual encouragement and assistance throughout the project work**.** We greatly appreciate the freedom and collegial respect. We are grateful to him for discussions about the technical matters and suggestions concerned to our topic.

We extend our sense of gratitude to **Dr. Nagaraj R Aiholli, Project Coordinator, Dept. of E&CE, JCE, Belagavi,** for extending support and cooperation which helped us in completion of the project work.

We extend our sense of gratitude to **Prof. V R Bagali, Head, Dept. of E&CE, JCE, Belagavi,** for extending support and cooperation which helped us in completion of the project work.

We would like to express our sincere thanks to **Dr. J Shivakumar, Principal and Director, JCE Belagavi,** for extending support and cooperation which helped us in the completion of the project work.

We would like to extend our gratitude to all staff of **Department of Electronics and Communication Engineering** for the help and support rendered to us.

We would like to extend our gratitude to all our family members and friends especially for their advice and moral support.

TABLE OF CONTENTS

Sl. No	Chapter description	Page. Number
1	Introduction	
1.1	Literature Survey	
1.2	Problem Statement	

TABLE OF CONTENTS(Should be in above format with page number with below contents shown chapterwise as reference format)

ABSTRACT

Smart farming is an emerging technology that combines IoT and Artificial Intelligence to improve agricultural productivity. In this project, a smart farming system is developed using an ESP32 microcontroller, soil moisture sensor, temperature and humidity sensor, and an AI/ML model for leaf disease classification. The ESP32 collects real-time environmental data from the sensors and transmits it wirelessly to a cloud-based IoT dashboard for monitoring. Additionally, leaf images captured from the field are used to train an AI/ML model that can identify plant diseases and provide early warnings to farmers.

The system enables continuous monitoring of soil and climate conditions, supports smart irrigation decision-making, and offers AI-based plant health analysis. The real-time dashboard allows farmers to remotely track crop status and take timely actions. This integrated IoT–AI solution enhances farm management, reduces manual effort, conserves resources, and improves crop yield. The project demonstrates that combining sensors, machine learning, and IoT technologies can create an efficient and intelligent agricultural support system.

CHAPTER 1. INTRODUCTION

1.1 Motivation

Agriculture is the backbone of India's economy, yet farmers face challenges such as labour shortages, inefficient resource usage, unpredictable weather, and crop management difficulties. With the rapid development of Artificial Intelligence (AI), Machine Learning (ML), Internet of Things (IoT), and robotics, there is a huge opportunity to modernize farming. The motivation behind this project is to design an intelligent agricultural robot that reduces manual labour, increases crop yield, and supports farmers with real-time decision-making.

1.2 Problem Statement

Traditional farming methods require continuous manual labour for tasks such as soil monitoring, plant health diagnosis, irrigation, and fertilizer application. These processes are time-consuming, costly, and prone to human error.
Thus, there is a need for an intelligent, autonomous agricultural robot that can monitor, analyze, and act on farm conditions in real time.

1.3 Objectives of the Project

1. To design an autonomous multipurpose robot capable of performing soil sensing, irrigation, and disease detection.
2. To integrate AI/ML algorithms for real-time plant disease detection using a mobile/embedded camera.
3. To provide centralized monitoring through an IoT dashboard.
4. To reduce farmer workload and improve precision in agriculture.
5. To ensure scalability and low operational cost for small-scale farmers.

1.4 Literature Survey

- Recent studies highlight the use of CNN models for leaf disease detection, achieving over 90% accuracy.
- IoT-based solutions have been widely used for soil moisture, temperature, humidity, and water level monitoring.
- Agricultural robots developed in previous research are often task-specific and lack multi-functionality.

- Integration of AI + IoT + Robotics is still emerging and has high potential for improving productivity.

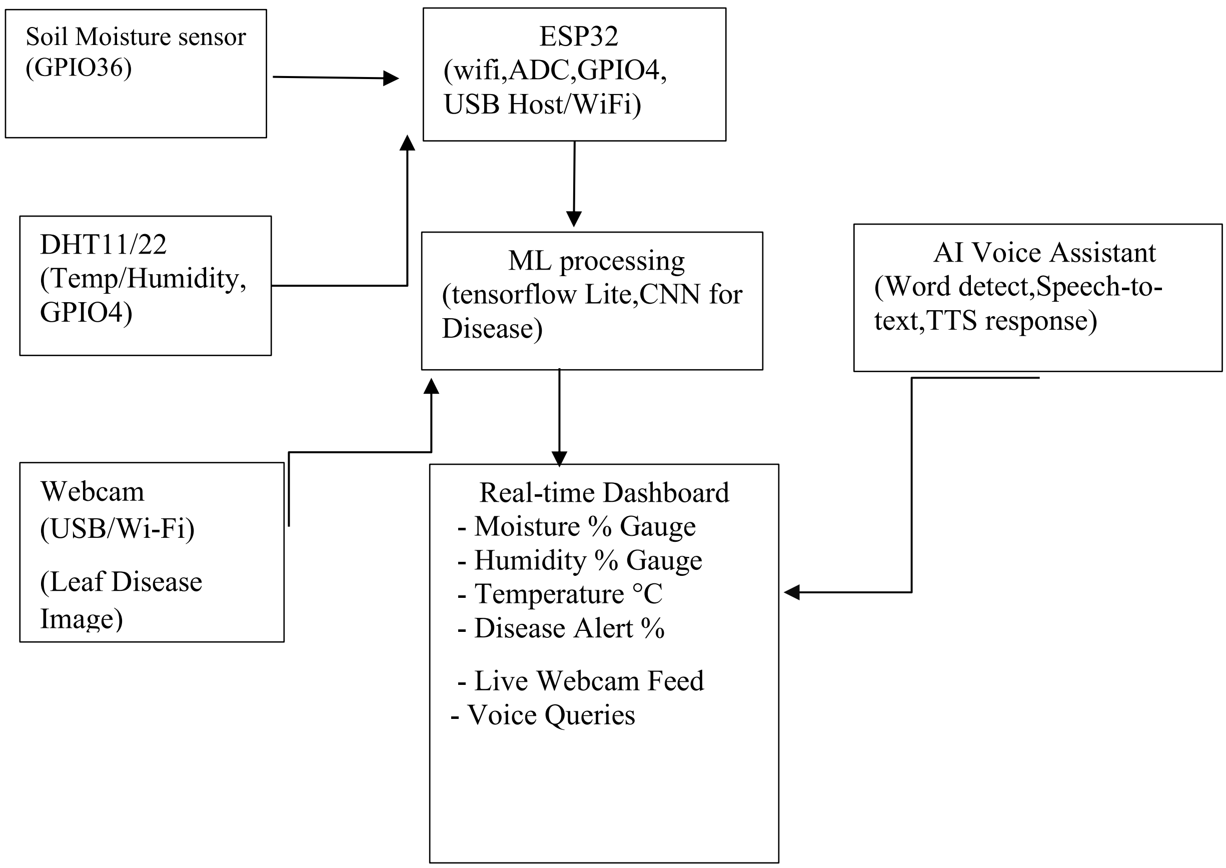
1.5 Design Approach

The AgriBot uses:

- **Sensors** (Moisture, DHT11/DHT22, pH sensor, ultrasonic sensors) for field data.
- **Camera module** (ESP32-CAM / mobile camera) for plant disease detection.
- **Microcontrollers** like Arduino and ESP32 for real-time control.
- **AI models** (CNN-based) deployed on server/cloud/mobile for disease classification.
- **IoT Dashboard** for live analytics and automated decision-making.

CHAPTER 2. HARDWARE DESIGN

Block Diagram



Soil Moisture and DHT11/22 Sensors

- The soil moisture sensor connects to an analog pin (GPIO36) of the ESP32 to measure soil water content as a variable voltage level.
- The DHT11/22 sensor connects to a digital GPIO (GPIO4) and provides temperature and humidity readings as digital data that the ESP32 decodes.

ESP32 Core Module

- The ESP32 block represents the main microcontroller with WiFi, ADC (for analog soil moisture), and digital GPIOs (for DHT11/22 and other peripherals).
- It collects sensor data, communicates with the ML processing block, sends/receives data from the AI voice assistant, and pushes all relevant information to the real-time dashboard over WiFi.

Webcam and ML Processing

- The webcam (USB or WiFi) captures images of plant leaves and sends them to the ESP32 or a connected ML processing unit.
- The "ML Processing" block runs a CNN-based model (e.g., with Edge Impulse or TensorFlow Lite) to analyze these images and classify leaf diseases, then outputs a disease status or probability back to the ESP32.

AI Voice Assistant

- The microphone (I2S) provides audio input to the ESP32 or a dedicated voice-processing module for speech recognition.
- The "AI Voice Assistant" block handles speech-to-text, command recognition (e.g., "show moisture level"), and text-to-speech responses, exchanging commands and status with the ESP32 and dashboard.

Real-Time Dashboard

- The "Real-time Dashboard" block (e.g., Blynk/ThingSpeak or a custom web UI) displays live values: soil moisture percentage, humidity, temperature, disease alerts, webcam feed, and processed voice commands.
- The dashboard both receives data from the ESP32 (sensor readings and disease results) and can send control actions or queries triggered by the voice assistant back to the ESP32 for execution.

.

CHAPTER 3: SOFTWARE DESIGN

3.1 Hardware Resource Allocation

Component	Connected To	Purpose
Soi Moisture Sensor	Analog Pin A0	Irrigation control
DHT11/DHT22 Sensor	Digital Pin D2	Temperature & humidity
ESP32	Separate 3.3V module	Leaf image capture
IoT Module	Built-in WiFi	Dashboard communication

3.2 Programming Language

1. EMBEDDED C

Used for Arduino/ESP32 programming.
Includes sensor reading, motor control, and relay operations.

2. PYTHON

Used for AI/ML model training for leaf disease detection (CNN model).
Libraries: TensorFlow, Keras, OpenCV, NumPy.

3. JAVASCRIPT / HTML

Used when developing a custom web dashboard for IoT monitoring.

Development Tools

ARDUINO IDE

Used to write, compile, and upload code to Arduino/ESP32.

VS CODE

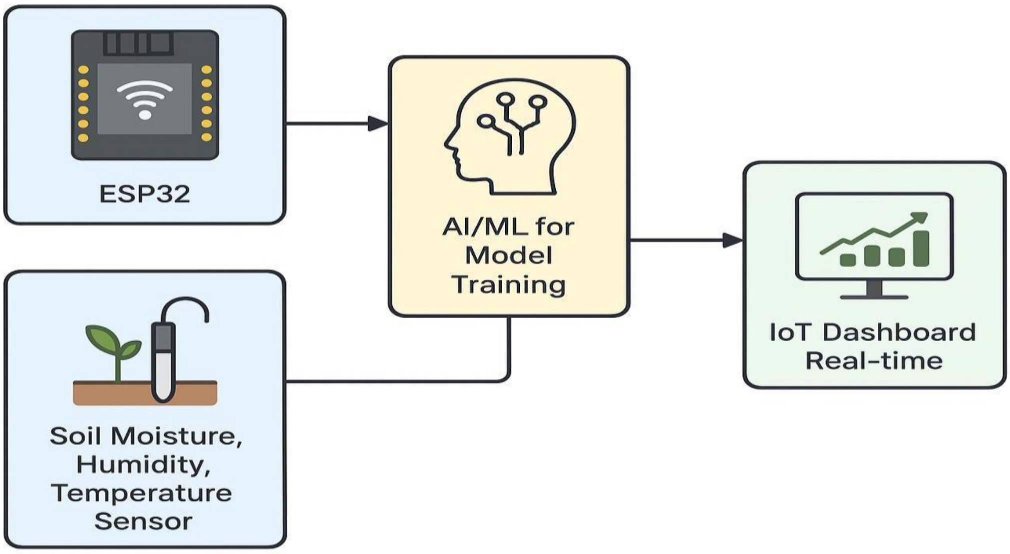
Used for AI model training, Python backend scripts, and debugging.

Mobile Apps / IoT Platforms

- Firebase
- MQTT Cloud

Used for real-time dashboard & notifications.

3.4 SOFTWARE DESIGN -DISCRIPTION



1. ESP32 Module

- The diagram starts with an **ESP32**, which acts as the main controller.
- It collects data from sensors and sends it to cloud/AI processing.
- Also used for wireless communication (Wi-Fi).

2. Soil Moisture, Humidity & Temperature Sensor

- This block shows combined environmental sensors.
- These sensors measure:
  - **Soil Moisture**
  - **Air Humidity**
  - **Temperature**
- The sensed values are sent directly to the ESP32 for processing.

3. AI/ML Model Training (Leaf Disease Detection)

- This block represents the **AI/ML system used for training models**, especially:
  - **Leaf image classification**
  - **Leaf disease detection**
  - **Stress detection (color, texture patterns)**
- Sensor data + leaf images from ESP32/ESP32-CAM are used to train the ML model.

4. IoT Dashboard (Real-time Monitoring)

- The final block shows a **real-time IoT dashboard**.
- It displays:
  - Soil moisture levels
  - Temperature & humidity
  - AI-based predictions about leaf health
  - Alerts or recommendations
- This helps farmers monitor crops remotely.

CHAPTER 4 APPLICATIONS

1. Smart Irrigation Control

- Automatically waters plants based on **soil moisture readings**.
- Helps avoid **over-irrigation** and **under-irrigation**.
- Conserves water and increases crop yield.

2. Real-time Crop Monitoring

- The IoT dashboard shows:
  - Soil moisture
  - Humidity & temperature
  - Leaf health status
- Farmers can monitor crops from anywhere using a mobile/PC dashboard.

3. Leaf Disease Detection (AI/ML)

- The AI/ML model analyzes leaf images to detect:
  - Fungal diseases
  - Bacterial spots
  - Nutrient deficiency
  - Pest infection
- Helps farmers take action early.

4. Climate Monitoring in Farms

- Sensors track environmental conditions:
  - Temperature
  - Humidity
- Helps to maintain the ideal climate for plant growth (greenhouses also).

CHAPTER 5 RESULTS

The smart farming system was successfully designed and implemented using **ESP32**, **soil moisture sensor**, **temperature-humidity sensor**, **AI/ML model**, and a **real-time IoT dashboard**.
The system accurately monitored environmental conditions such as soil moisture, temperature, and humidity, and transmitted the data to the IoT dashboard in real time.

The AI/ML model effectively analyzed leaf images and provided predictions for **leaf disease detection**, helping in early identification of plant health issues.

The system demonstrated:

- **Accurate sensor readings**
- **Stable wireless data transmission** through ESP32
- **Successful visualization** on the IoT dashboard
- **Correct AI predictions** during leaf detection tests

5.1 MODULE TESTING

- ### 5.1.1 Power Testing

  The power module was tested using a regulated 5V supply for the ESP32 and sensors.
  The ESP32 successfully powered ON with stable operation. Voltage was measured at different points and remained within safe operating limits.

  Result:

- ESP32 boot stable
- Sensors receiving proper 5V
- No overheating or voltage drop

- ### 5.1.2 DHT11 Sensor Testing

  The DHT11 temperature and humidity sensor was connected to the ESP32 GPIO pin and tested.
  Sensor readings were displayed on the serial monitor.

  Observations:

- Temperature reading responded correctly to warm/cool air

- Humidity increased when moisture sources were brought nearby

Result:
DHT11 sensor functioning accurately with stable readings.

- **5.1.3 Soil Moisture Sensor Testing**

The soil moisture sensor (analog type) was tested with dry soil, moist soil, and wet soil.

Results:

Soil Condition	Sensor Output	Interpretation
Dry Soil	High value	Low moisture
Moist Soil	Mid value	Medium moisture
Wet Soil	Low value	High moisture

The sensor values changed consistently based on soil wetness.

- **5.1.4 ESP32 Board Testing**

The ESP32 microcontroller was tested for serial communication, WiFi connectivity, and GPIO operations.

Results:

- WiFi successfully connected to hotspot/network
- Sensor readings displayed correctly
- GPIO pins responded accurately

- **5.1.5 IoT Dashboard Testing**

The IoT dashboard (Blynk / Thingspeak / Firebase) was tested for real-time data updates.

Result:

- Sensor readings uploaded successfully
- Dashboard refreshed values every 2–5 seconds

- No data loss or communication failure observed

- **5.2 SYSTEM TESTING**

  System testing involved combining all modules and verifying that they operate correctly as a complete Smart Agribot system.

- **5.2.1 ESP32 with DHT11 Sensor**

  The ESP32 and DHT11 were integrated and tested.
  Values of temperature and humidity were displayed both locally and via the IoT dashboard.

  Result:

- Stable temperature/humidity monitoring
- Data accurately transmitted to cloud dashboard

- **5.2.2 ESP32 with Soil Moisture Sensor**

  The ESP32 and soil moisture sensor were integrated and tested in various soil types.

  Testing observations:

- In dry soil → IoT shows LOW MOISTURE
- In moist soil → IoT shows NORMAL MOISTURE
- In wet soil → IoT shows HIGH MOISTURE

  Result:
  Accurate soil moisture detection with reliable updates on IoT platform.

- **5.2.3 ESP32 with Both Sensors (DHT11 + Soil Moisture)**

  Both sensors were connected simultaneously and tested for combined performance.

  Result:

- No interference between sensors
- ESP32 handled analog + digital readings smoothly

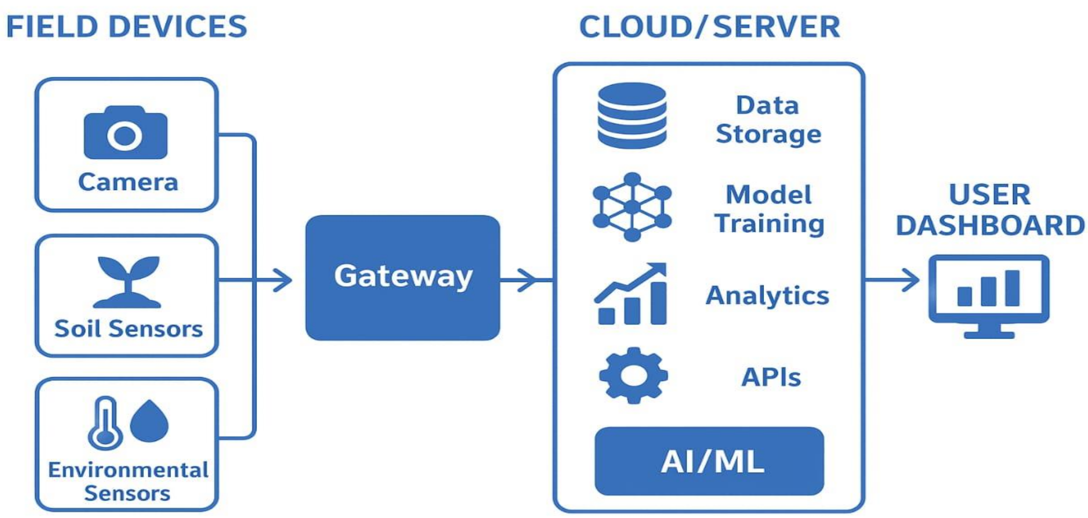
- IoT dashboard displayed all values in real time

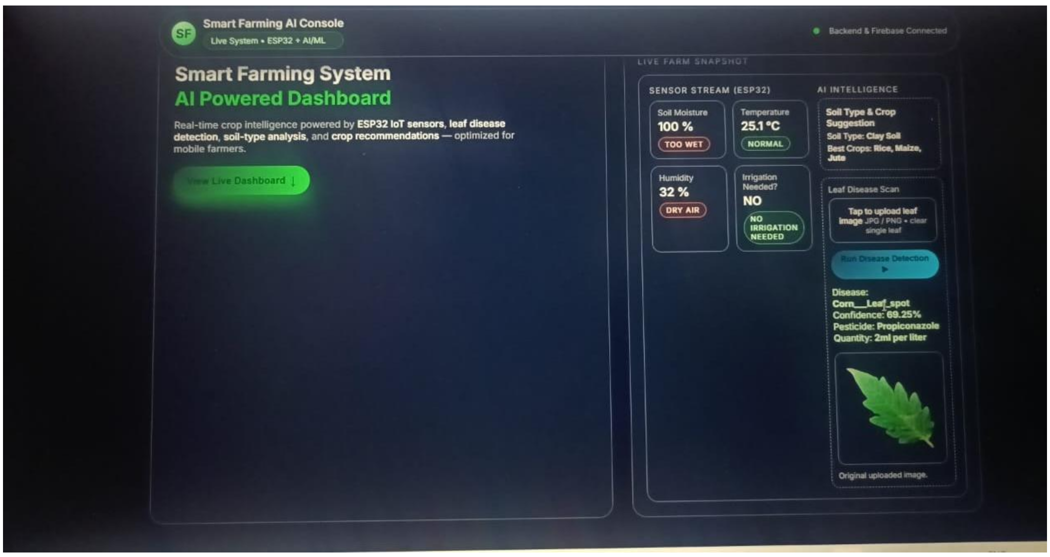
- **5.2.4 ESP32 with IoT Cloud**

  ESP32 was connected to WiFi and data was pushed to the cloud platform.

  Result:

- Real-time monitoring achieved
- Dashboard inserts and updates were accurate
- Cloud-based history logs generated correctly





CHAPTER 6 CONCLUSION

The smart farming system developed using **ESP32**, **soil moisture sensor**, **temperature–humidity sensor**, **AI/ML leaf detection**, and a **real-time IoT dashboard** successfully demonstrates how modern technology can improve agricultural practices. The project proved that real-time monitoring of soil and environmental conditions helps in making accurate irrigation decisions, reducing water usage, and improving crop health.

The AI/ML model effectively identified leaf diseases and provided early alerts, enabling farmers to take preventive measures before crop damage occurs. The IoT dashboard allowed continuous and remote supervision of farm conditions, making the system efficient and user-friendly.

Overall, the project shows that integrating **IoT + AI/ML** in agriculture can enhance productivity, reduce manual labor, improve decision-making, and support sustainable farming. This solution is scalable, cost-effective, and can be extended to large farms or greenhouses in the future.

CHAPTER 6 FUTURE SCOPE

Hardware Upgrades (Low Cost)

- Replace DHT11 with DHT22 or SHT-series only if higher accuracy is needed, keeping the basic ESP32 + soil moisture + single webcam architecture to avoid extra controllers.
- Add a few more low-cost sensors (light/UV sensor, simple rain sensor) so the same ESP32 board gives more insights without major cost increase.

Connectivity and Power

- Introduce optional LoRa or WiFi-range extenders only for large farms, so small plots can still work with just WiFi and a cheap router.
- Use a small solar panel plus battery for remote fields to reduce wiring cost and electricity dependency.

Smarter but Lightweight AI

- Start with simple threshold-based rules (e.g., moisture < X) and gradually add lightweight models for better disease classification and irrigation decisions, keeping inference on-device to avoid cloud fees.

- Periodically retrain the disease model using farmer-captured images so performance improves over time without buying new hardware.

Affordable Dashboard Enhancements

- Keep the embedded web dashboard (hosted on ESP32) as the default so there is no recurring cloud subscription.
- Add export to CSV or simple offline logging on SD card, so farmers can view trends on any PC or mobile later instead of paying for advanced analytics tools.

Scalability for Farmer Groups

- Allow multiple ESP32 nodes to send data to a single low-cost central gateway (another ESP32/Raspberry Pi) so a group of farmers can share infrastructure.
- Provide simple configuration pages on the dashboard so non-technical users can add new nodes or change thresholds without needing a developer.

REFRENCES

1. D. Singh and B. Singh, "Plant disease detection using deep learning," *Machine Learning for Agriculture*, Springer, pp. 29–43, 2020.
2. T. Choudhary, A. Mishra, and S. Gupta, "AI-powered smart farming: A review," *Journal of Intelligent & Fuzzy Systems*, vol. 38, pp. 1–13, 2020
3. M. T. Islam, "Sensors in smart agriculture," *IEEE Sensors Journal*, vol. 21, no. 10, pp. 11540–11557, 2021.
4. TensorFlow, "TensorFlow Lite for Microcontrollers – Leaf Disease Classification Guide," Google AI, 2023.
5. **Espressif Systems.** *ESP32 Series Datasheet.* Espressif Systems, 2023. Available at: https://www.espressif.com (Accessed 2025).
6. **DHT11 Temperature & Humidity Sensor Datasheet.** Aosong Electronics Co., Ltd., 2022.
7. **YL-69 Soil Moisture Sensor Module Datasheet.** Shenzhen Longruner Technology, 2021.
8. **P. Rajalakshmi and S. Devi Mahalakshmi.** "IoT Based Crop-Field Monitoring and Irrigation Automation." *2016 IEEE Region 10 Conference (TENCON)*, IEEE, 2016.
9. **Z. Ahmed et al.** "Smart Agriculture Using IoT." *International Journal of Advanced Research in Computer Science*, vol. 8, no. 5, 2017, pp. 424–431.
10. **A. Tzounis et al.** "Internet of Things in Agriculture: Applications, Challenges and Opportunities." *Computers and Electronics in Agriculture*, vol. 163, Elsevier, 2019.
11. **Blynk IoT Platform Documentation.** Blynk IoT, 2024. Available at: https://docs.blynk.io
12. **Thingspeak IoT Cloud Documentation.** MathWorks, 2024. Available at: https://thingspeak.com
13. **Firebase Realtime Database Documentation.** Google Developers, 2024. Available at: https://firebase.google.com
14. **Aravindh, K. & Jayaprakash, M.** "Design and Implementation of a Smart Irrigation System Using IoT." *International Journal of Engineering and Technology*, vol. 7, 2018.

APPENDIX A-LIST OF COMPONENTS

ESP32 Development Board

- Microcontroller with Wi-Fi and Bluetooth
- Used for sensor interfacing and IoT connectivity

Soil Moisture Sensor

- Type: Capacitive or Resistive
- Measures water content in soil

DHT11 / DHT22 Humidity & Temperature Sensor

- Measures ambient temperature and humidity
- Used for climate monitoring

Jumper Wires (Male-Female, Male-Male)

- For circuit connections

Breadboard

- Used for prototyping sensor connections

USB Cable (Micro USB / Type-C)

- For ESP32 programming and power

Power Supply

- 5V DC adapter or power bank
- Provides stable power to ESP32 and sensors

Cloud IoT Platform / Dashboard

- Examples: Thingspeak, Blynk, Firebase, Node-RED
- Displays real-time farm data

AI/ML Model Resources

- Dataset of plant leaf images
- Training tools: TensorFlow, Python, Google Colab

APPENDIX B-PSEUDOCODE

```
START

// --- INITIALIZATION ---
Initialize ESP32 microcontroller
Initialize WiFi connection
Initialize DHT11 sensor (temperature & humidity)
Initialize Soil Moisture Sensor (analog input)

// --- MAIN LOOP ---
REPEAT FOREVER

  // Read temperature and humidity
  temp ← Read temperature from DHT11
  humidity ← Read humidity from DHT11

  // Read soil moisture level
  moistureValue ← Read analog value from soil moisture sensor

  // Convert moisture value into percentage
  moisturePercent ← Map sensor value to 0–100%

  // Display and send data
  Print temp, humidity, moisturePercent on serial monitor
  Send data to IoT Dashboard (Blynk / Firebase / Thingspeak)

  // Check moisture condition
  IF moisturePercent < 30 THEN
     Status ← "Soil Dry"
  ELSE
     Status ← "Soil Normal"
```

```
    ENDIF

    // Optional alert message
    IF temp > 40 THEN
        Send alert "High Temperature"
    ENDIF

    WAIT for 2 seconds

END REPEAT

END
```